

USTECH
INNOVATIONS

USTECH ProLine17ECS

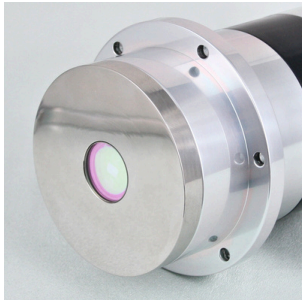
Continuous Analysis, Real-Time Quality



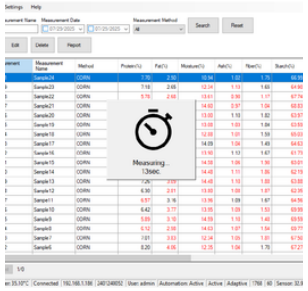
USTECH ProLine17ECS

Continuous Analysis, Real-Time Quality

USTECH ProLine17ESS is an advanced Near-Infrared (NIR) Spectroscopy system that allows you to monitor key parameters in your production process directly, continuously, and in mere seconds. Integrating seamlessly into the process line without requiring a bypass, it delivers complete traceability—maximizing your yield and profitability from raw material intake to final product approval.







High-Precision Spectroscopy

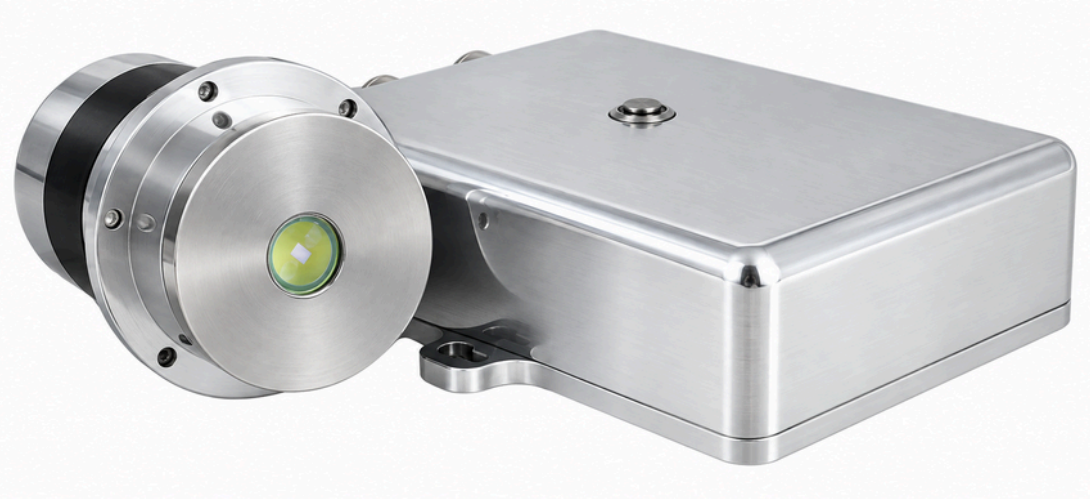
Equipped with an InGaAs photodiode array operating within the 950-1650 nm wavelength range. Delivering 2000 spectra per sample and a typical signal-to-noise ratio of 23,000:1 , it performs highly sensitive analyses that go well beyond industry standards.

Flawless Process Integration

Engineered with a 316 Stainless Steel housing and a 3 mm thick food-grade sapphire interface (O-ring sealed), it ensures absolute durability against harsh production conditions. Designed with an IP65 protection rating, it is easily mounted directly onto your process line.

Continuous & Instant Data Flow

Delivers analytical results in an average of 2 to 20 seconds (maximum <30 sec). By directly transmitting the acquired parameters to your control room systems, it empowers you with the capability for instantaneous operational intervention.



Technical Specifications

Function Range	0-(+35) °C
Ingress Protection	IP65
Dimensions(DxL)	96mm x 180mm
Material	Stainless Steel 316
Software	ProChem for Instrumental Control
Wavelength Range	950-1650 nm
Sample Type	Direct measurement of solid and liquid samples in ground or unground from (including pellets)
Parameters	Fat, protein, moisture, ash, starch and crude fiber

Industry Applications



Food & Feed

Optimizes your yield across the entire value chain, from raw material intake to final product release. It instantly detects critical nutritional parameters—such as fat, protein, moisture, and starch—across diverse sample forms (solid, liquid, or pellets) with absolutely zero preparation required.



Pharmaceutical & Chemical

Ensures strict regulatory compliance and batch uniformity. Leveraging non-destructive technology, it delivers highly precise measurements of Active Pharmaceutical Ingredients (API) and complex reaction dynamics directly within the process line, eliminating the risk of cross-contamination and hazardous sampling.



Bioenergy & Biomass

Maximizes process efficiency and enables sustainable energy production. Real-time NIR analysis allows continuous monitoring of quality variations in biomass feedstock. Instant control of key parameters such as moisture, ash, cellulose, and others ensures optimized fuel quality, improved energy efficiency, and reduced operational costs.

Our Core Commitments to You

"The Standard in Spectroscopy" is our commitment to your success. We provide cutting-edge analytical solutions so you can focus entirely on optimizing production, streamlining workflows, and ensuring uncompromised product quality.



Innovative

We leverage advanced technological expertise to develop state-of-the-art solutions—like direct measurement systems—that redefine laboratory standards.



Reliable

We guarantee precision and reproducibility, ensuring consistent, lab-grade results you can trust in every analysis.



Effortless

We simplify your daily workflow with intuitively designed instruments requiring zero-to-minimal sample preparation.



Cost-Effective

Maximize economic benefit by completely eliminating the need for expensive consumables, toxic reagents, and time-consuming methods.



Safe & Clean

We prioritize operator safety and the environment through robust housing architectures and chemical-free analytical processes.



Fast

Empowering rapid decisions with instant insights. We deliver accurate results in under a minute to keep your production moving.

CONTACT US

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